Text Based Analysis of Emerging Risks

From Cyber Risk to Multi-Risk Classification

Giovanni Della Lunga

University of Bologna

June 19, 2025

Overview

- **Current Focus**: Text-based measures of cyber risk using news attention indices
- **Methodology**: Natural language processing applied to financial risk assessment
- **Future Development**: Extension to multi-label classification of emerging risks
- **Key Innovation**: From single-risk attention indices to comprehensive risk taxonomy

Research Question

Can text-based analysis effectively identify and classify multiple emerging risks (climate, cyber, geopolitical) in corporate disclosures and news media?

U. Cherubini (University of Bologna) - G. Della Lunga (University of Bologna)- G. Figà Talamanca (University of Perugia) - B. Guardabascio (University of Perugia) *Text-Based Measures of Cyber Risk, The Economic Impact of Cyber Risk News*. Work in progress.

Current Methodology: Cyber News Attention Index (CNAI)

Data Sources:

- 5 major newspapers (2014-2023)
- Financial Times, LA Times, NY Times, USA Today, WSJ
- 10-year temporal horizon

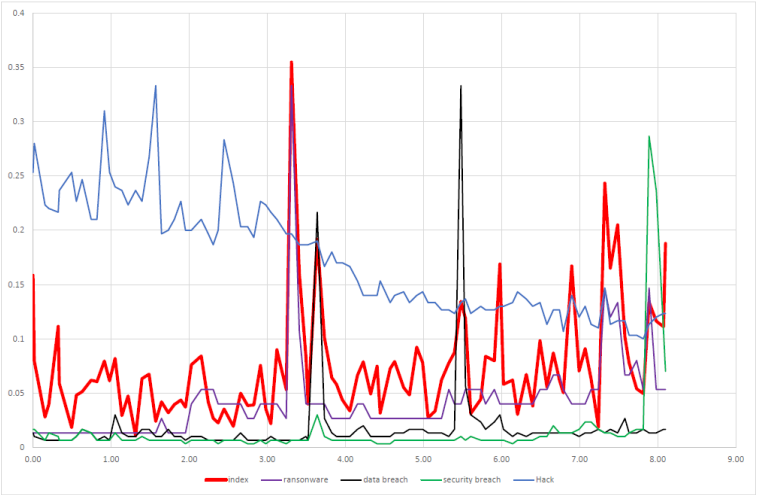
Processing Pipeline:

1. Text preprocessing (tokenization, stopword removal)
2. Keyword-based scoring (0-10 scale)
3. Daily/weekly/monthly aggregation
4. Validation with Google Trends

Keywords Examples

- malware
- phishing
- data breach
- ransomware
- unauthorized access

Current Methodology: Cyber News Attention Index (CNAI)



CNAI Validation: Major Cyber Events Detected

Date	Event
May 2017	WannaCry Ransomware (200K+ computers)
Sep 2017	Equifax Breach (143M people affected)
Dec 2020	SolarWinds Hack (US gov agencies)
May 2021	Colonial Pipeline Ransomware
Dec 2021	Log4j Vulnerability

- **18 "months of attention"** identified (threshold-based)
- Strong correlation with Google Trends data
- External validation through ChatGPT event identification

Empirical Analysis: Market Impact Assessment

Mimicking Portfolio Approach:

$$\Delta CNAI_t = \alpha + \beta Z_t^{Cyb} + \gamma' Z_t + \omega_t$$

Where $Z^{Cyb} \equiv R'(I_A - I_{NA})$

- Long-short portfolio
- Attacked vs. non-attacked firms
- Controls: market indices, size, value factors

Local Projection Analysis:

$$R_{i,t+k} = \alpha_k + \beta_k \Delta CNAI_t + \gamma_k' Z_t + \epsilon_{t+k}$$

- Impulse response functions
- Sector-level analysis (GICS)
- Multi-horizon effects ($k = 1, ..., n$)

Firm-Based Data

- In the United States, publicly traded companies are required to periodically submit documents to the Securities and Exchange Commission (SEC) to ensure transparency and investor protection.
- The main disclosure forms include:
  - Form 10-K (annual report)
  - Form 10-Q (quarterly report)
  - Form 8-K (current report for material events)

Form 10-K

- **Definition:** Form 10-K is the annual report that publicly traded companies must submit to the SEC, providing a comprehensive overview of their financial performance.
- **Content:**
  - Company history and organizational structure.
  - Audited financial statements.
  - Analysis of economic-financial results.
  - Description of business strategy.
  - Discussion of risk factors.
  - Governance information.
- **Filing deadlines:**
  - Large Accelerated Filers (market capitalization $\geq$ 700 million USD): within 60 days of fiscal year end.
  - Accelerated Filers (market capitalization between 75 and 700 million USD): within 75 days.
  - Non-Accelerated Filers (market capitalization ¡ 75 million USD): within 90 days.

Form 10-K



UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-K

(Mark One)

☒ ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934
For the fiscal year ended September 28, 2024
or
☐ TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934
For the transition period from _____ to _____
Commission File Number: 001-36743

Apple Inc.
(Exact name of Registrant as specified in its charter)

California	94-2404110
(State or other jurisdiction of incorporation or organization)	(I.R.S. Employer Identification No.)
One Apple Park Way Cupertino, California	95014
(Address of principal executive offices)	(Zip Code)

(408) 996-1010
(Registrant's telephone number, including area code)

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading symbol(s)	Name of each exchange on which registered
Common Stock, $0.00001 par value per share	AAPL	The Nasdaq Stock Market LLC
0.000% Notes due 2025	—	The Nasdaq Stock Market LLC
0.875% Notes due 2030	—	The Nasdaq Stock Market LLC
1.625% Notes due 2026	—	The Nasdaq Stock Market LLC
1.625% Notes due 2035	—	The Nasdaq Stock Market LLC
1.375% Notes due 2029	—	The Nasdaq Stock Market LLC
3.050% Notes due 2029	—	The Nasdaq Stock Market LLC
3.500% Notes due 2031	—	The Nasdaq Stock Market LLC
3.600% Notes due 2042	—	The Nasdaq Stock Market LLC

Securities registered pursuant to Section 12(g) of the Act: None

The Company is also subject to the examination of its tax returns and other tax matters by the U.S. Internal Revenue Service and other tax authorities and governmental bodies. The Company regularly assesses the likelihood of an adverse outcome resulting from these examinations to determine the adequacy of its provision for taxes. There can be no assurance as to the outcome of these examinations. If the Company's effective tax rates were to increase, or if the ultimate determination of the Company's taxes owed is for an amount in excess of amounts previously accrued, the Company's business, results of operations and financial condition could be materially adversely affected.

General Risks

The price of the Company's stock is subject to volatility.

The Company's stock has experienced substantial price volatility in the past and may continue to do so in the future. Additionally, the Company, the technology industry and the stock market as a whole have, from time to time, experienced extreme stock price and volume fluctuations that have affected stock prices in ways that may have been unrelated to those companies' operating performance. Price volatility may cause the average price at which the Company repurchases its stock in a given period to exceed the stock's price at a given point in time. The Company believes the price of its stock should reflect expectations of future growth and profitability. The Company also believes the price of its stock should reflect expectations that its cash dividend will continue at current levels or grow, and that its current share repurchase program will be completed as planned or at all. Consequently, if the Company's growth or profitability slows, if the Company announces changes in its business, operations or strategy, or if the Company fails to meet expectations related to future growth, profitability, dividends, share repurchases or other market expectations, the price of the Company's stock may decline significantly, which could have a material adverse impact on investor confidence and employee retention.

Item 1B. Unresolved Staff Comments

None.

Item 1C. Cybersecurity

The Company's management, led by its Head of Corporate Information Security, has overall responsibility for identifying, assessing and managing any material risks from cybersecurity threats. The Company's Head of Corporate Information Security leads a dedicated Information Security team of highly skilled individuals with experience across industries that, among other things, develops and distributes information security policies, standards and procedures; engages in employee cybersecurity training; implements security controls; assesses security risk and vulnerabilities; and monitors and tests the Company's systems on an ongoing basis. The Company's Head of Corporate Information Security has extensive knowledge and skills gained from over 25 years of experience in the cybersecurity industry, including serving in leadership positions at other large technology companies and leading the Company's Information Security team since 2015.

The Company's Information Security team coordinates with teams across the Company to prevent, respond to and manage security incidents, and engages third parties, as appropriate, to assess, test or otherwise assist with aspects of its security processes and incident response. A dedicated Supplier Trust team manages information security risks the Company is exposed to through its supplier relationships. The Company has processes in log, track, address, and escalate for further assessment and report, as appropriate, cybersecurity incidents across the Company and its suppliers to senior management and the Audit and Finance Committee (the "Audit Committee") of the Board. The Company's enterprise risk management program is designed to identify, assess, and monitor the Company's business risks, including financial, operational, compliance and reputational risks, and reflects management's assessment of cybersecurity risks.

The Audit Committee assists the Board in the oversight and monitoring of cybersecurity matters. The Audit Committee regularly reviews and discusses the Company's cybersecurity risks with management, including the Company's Head of Corporate Information Security, its General Counsel and the Heads of Compliance and Business Conduct, Business Assurance, and Internal Audit, and receives updates, as necessary, regarding cybersecurity incidents. The Chair of the Audit Committee regularly reports the substance of such reviews and discussions to the Board, as necessary, and recommends to the Board such actions as the Audit Committee deems appropriate.

For a discussion of the Company's cybersecurity-related risks, see Item 1A of this Form 10-K under the heading "Risk Factors."

Fonte: Form 10-K AAPL (2024)

Form 8-K

- **Definition:** Form 8-K is a current report used to communicate material corporate events that occur between periodic filings.
- **Examples of events to report:**
  - Changes in corporate leadership.
  - Significant acquisitions or divestitures.
  - Bankruptcies.
  - Material cybersecurity incidents.
- **Filing timeline:** Companies must file Form 8-K within 4 business days of the event (with possible extension up to 60 days).

Form 8-K



UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
WASHINGTON, D.C. 20549

FORM 8-K

CURRENT REPORT
PURSUANT TO SECTION 13 OR 15(D)
OF THE SECURITIES EXCHANGE ACT OF 1934

Date of Report (Date of earliest event reported): January 17, 2024

Microsoft Corporation

Washington	001-37845	91-1144442
(State or other jurisdiction of incorporation)	(Commission File Number)	(IRS Employer Identification No.)

One Microsoft Way, Redmond, Washington 98052-6399

(425) 882-8080
www.microsoft.com/investor

Check the appropriate box below if the Form 8-K filing is intended to simultaneously satisfy the filing obligation of the registrant under any of the following provisions (see General Instruction A.2. below):

☐ Written communications pursuant to Rule 425 under the Securities Act (17 CFR 230.425)
☐ Soliciting material pursuant to Rule 14a-12 under the Exchange Act (17 CFR 240.14a-12)
☐ Pre-commencement communications pursuant to Rule 14d-2(b) under the Exchange Act (17 CFR 240.14d-2(b))
☐ Pre-commencement communications pursuant to Rule 13e-4(c) under the Exchange Act (17 CFR 240.13e-4(c))

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol	Name of exchange on which registered
Common stock, $0.00000625 par value per share	MSFT	NASDAQ
3.125% Notes due 2028	MSFT	NASDAQ
2.625% Notes due 2033	MSFT	NASDAQ

Indicate by check mark whether the registrant is an emerging growth company as defined in Rule 405 of the Securities Act of 1933 (§230.405 of this chapter) or Rule 12b-2 of the Securities Exchange Act of 1934 (§240.12b-2 of this chapter). Emerging growth company ☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Item 1.05. Material Cybersecurity Incidents

On January 12, 2024, Microsoft (the "Company" or "we") detected that beginning in late November 2023, a nation-state associated threat actor had gained unauthorized access to some of the Company's corporate email accounts including members of our senior leadership team and employees in our cybersecurity, legal, and other functions, on the basis of preliminary analysis. We were able to remove the threat actor's access to the email accounts on or about January 13, 2024. We are examining the information accessed to determine the impact of the incident. We also continue to investigate the extent of the incident. We have notified and are working with law enforcement. We are also notifying relevant regulatory authorities and will be notifying impacted customers as appropriate. To date, the incident has not had a material impact on the Company's financial condition or operations. The Company has not yet determined whether the incident is reasonably likely to materially impact the Company's financial condition or results of operations.

This Form 8-K contains forward-looking statements, which are any predictions, projections or other statements about future events based on current expectations and assumptions that are subject to risks and uncertainties, which are described in our filings with the Securities and Exchange Commission. Forward-looking statements speak only as of the date they are made. Readers are cautioned not to put undue reliance on forward-looking statements, and the Company undertakes no duty to update any forward-looking statement to conform the statement to actual results or changes in the Company's expectations.

Item 7.01. Regulation FD Disclosure

On January 19, 2024, the Company posted a blog regarding the incident. A copy of the blog is furnished as Exhibit 99.1 to this report.

In accordance with General Instruction B.2 of Form 8-K, the information in this Item 7.01 and Exhibit 99.1, shall not be deemed to be "filed" for purposes of Section 18 of the Securities Exchange Act of 1934, as amended (the "Exchange Act"), or otherwise subject to the liability of that section, and shall not be incorporated by reference into any registration statement or other document filed under the Securities Act of 1933, as amended, or the Exchange Act, except as shall be expressly set forth by specific reference in such filing.

Item 9.01. Financial Statements and Exhibits

(d) Exhibits.

99.1	Microsoft Blog Post dated January 19, 2024 titled "Microsoft Actions Following Attack by Nation State Actor Midnight Blizzard"
104	Cover Page Interactive Data File (embedded within the Inline XBRL document)

Fonte: Form 8-K MSFT (17/01/2024)

Data Integration: SEC 8-K Forms Analysis

Regulatory Framework (Post-2023):

- Mandatory cyber incident disclosure (Item 1.05)
- 4-day filing requirement
- Material cybersecurity events

Processing Pipeline:

1. Python + edgartools library for EDGAR access
2. Keyword-based filtering (20% threshold)
3. GPT-4 classification of attack types
4. Entity extraction (organizations, individuals, locations)
5. PostgreSQL storage for structured analysis

Limitation: Disclosure mandatory only from 2023 $\rightarrow$ retrospective vulnerability assumption

BERT for Document Classification

- **What is BERT?**
  - Bidirectional Encoder Representations from Transformers (Google, 2018)
  - Pre-trained language model that understands context from both directions
  - Captures deep semantic meaning and relationships in text
- **How BERT Works for Classification:**
  - **Input:** Document text is tokenized and fed to BERT with special [CLS] token
  - **Processing:** BERT generates contextualized embeddings for each token
  - **Classification:** [CLS] token representation is used for final classification layer
  - **Fine-tuning:** Model is adapted to specific classification tasks with labeled data

BERT for Document Classification

- **Applications for Document Classification:**
  - Sentiment analysis (positive/negative/neutral)
  - Topic classification (finance, technology, healthcare, etc.)
  - Document type identification (reports, emails, contracts)
  - Risk assessment and compliance categorization
- **Advantages:**
  - High accuracy due to contextual understanding
  - Transfer learning capabilities from pre-trained models
  - Handles complex linguistic patterns and domain-specific terminology

Future Development: Multi-Risk BERT Classification

Objective

Extend methodology to classify firms into multiple emerging risk categories using 10-K and 8-K disclosures

Target Risk Categories:

- Climate Physical Risk
- Climate Transition Risk
- Cybersecurity Risk
- Geopolitical Risk
- Supply Chain Risk
- Regulatory Risk
- Pandemic Risk

Data Source: Item 1A. Risk Factors section of 10-K filings

Advanced Methodological Approaches

1. **Weak Supervision (Snorkel):**
   - Labeling functions for scalability
   - Keyword-based heuristics
   - Probabilistic label generation
   - Reduced manual annotation effort
2. **Zero-Shot Classification:**
   - Natural Language Inference models
   - BART-large-mnli, RoBERTa-base
   - No training data required
   - Flexible for emerging risk types

3. **Hybrid Strategy:**
   1. Zero-shot for initial classification
   2. Generate pseudo-labels
   3. Train Snorkel labeling functions
   4. Fine-tune BERT on combined data

Advantage

Combines rapid prototyping with scalable, structured control over labeling process

What is Snorkel and Weak Labeling?

The Problem:

- Manual data labeling is **expensive** and **time-consuming**
- Need thousands of labeled examples to train ML models
- Expert annotators are costly and may disagree

Snorkel's Solution: Weak Supervision

1. **Labeling Functions**: Simple rules that assign labels
   - Example: "If text contains 'data breach' $\rightarrow$ Cyber Risk"
   - Example: "If text mentions 'carbon emissions' $\rightarrow$ Climate Risk"
2. **Multiple Noisy Sources**:
   - Keyword matching
   - Regular expressions
   - External databases
   - Existing models
3. **Statistical Combination**: Snorkel learns source reliability and combines them

Key Advantage: Label thousands of documents automatically with reasonable accuracy

What is Zero-Shot Learning?

Traditional Machine Learning:

$$\text{Training Data} \rightarrow \text{Train Model} \rightarrow \text{Predict on New Data}$$

Need labeled examples for each category

Zero-Shot Learning:

$$\text{Pre-trained Model} + \text{Task Description} \rightarrow \text{Direct Prediction}$$

No training examples needed!

How it Works:

- Use large language models (BART, RoBERTa) trained on massive text
- Provide natural language descriptions of categories:
  - "This text discusses cybersecurity threats and data breaches"
  - "This text covers climate change and environmental risks"
- Model uses learned language understanding to classify

Combining Snorkel and Zero-Shot Learning

Hybrid Strategy - Best of Both Worlds:

1. **Start with Zero-Shot**
   - Quickly classify large dataset
   - No manual labeling required
   - Generate initial pseudo-labels
2. **Create Snorkel Labeling Functions**
   - Use zero-shot results to identify patterns
   - Write keyword-based rules
   - Add domain expertise through heuristics
3. **Combine and Refine**
   - Snorkel weighs zero-shot + rule-based labels
   - Statistical model resolves conflicts
   - Produces higher-quality training data

Conclusion

Current Achievements:

- Successful CNAI development with strong external validation
- Integration of news-based and regulatory disclosure data
- Theoretical framework connecting text measures to asset pricing

Next Steps:

- Scale from single-risk (cyber) to multi-risk classification
- Implement BERT-based multi-label approach with weak supervision
- Develop comprehensive emerging risk taxonomy
- Create tradable risk indices for portfolio applications

Thank you for your attention!